General Goal

The objective of this project is to examine the understanding of the students in content taught by this course, by developing an AI-based computer vision solution for automated pick and place operation.

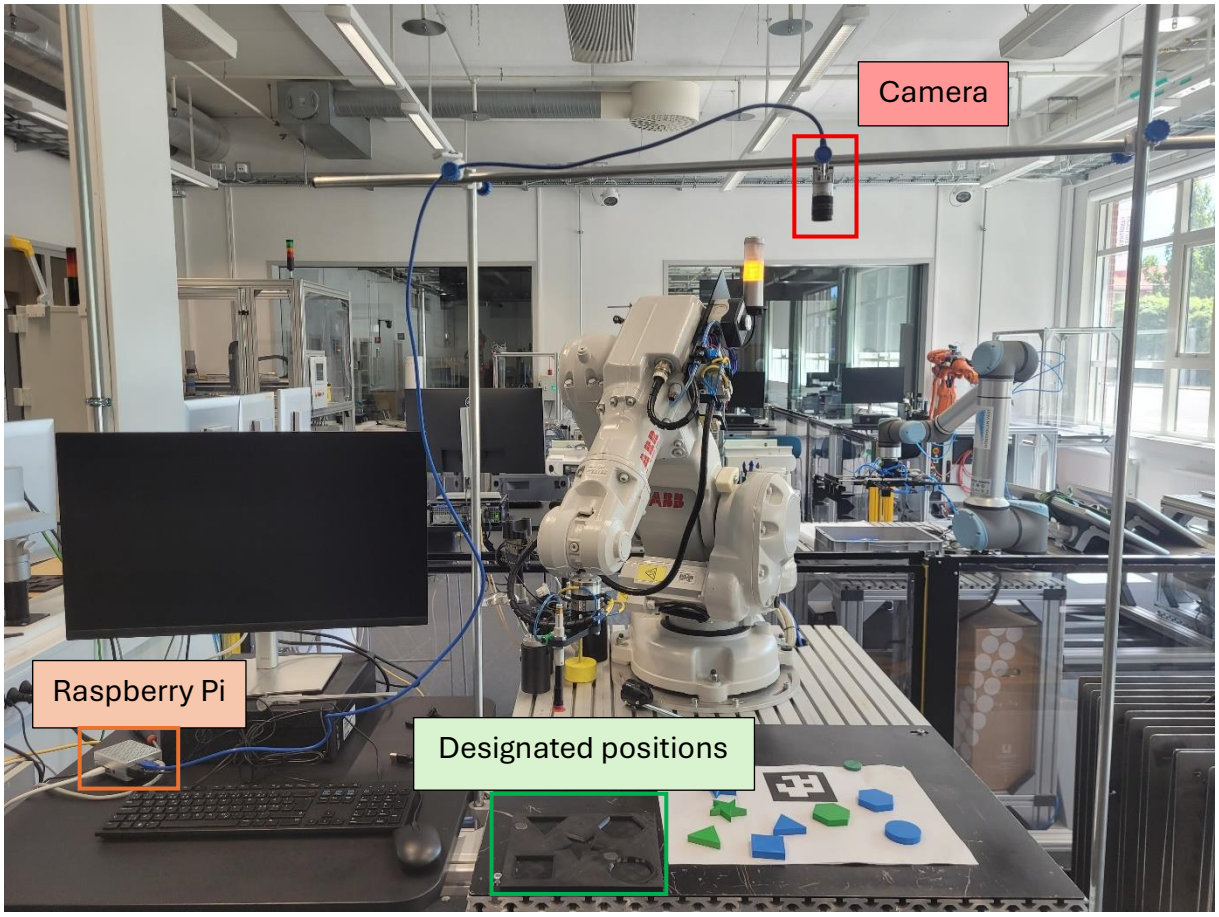


Figure 1 Robot station with Raspberry pi and camera

Project Description

In this project, a vision system will be implemented on a Raspberry Pi 4. The system will be responsible for detecting and classifying **three out of five different object shapes** placed in front of a robot, figure 2. An AI model will be trained to recognize the **shapes**, and to extract both the **position** and **orientation** of each object.

Once detected, the object's **pixel coordinates will be converted into the robot's work object coordinates**, allowing the robot to move accurately to the correct picking position. This data will then be transmitted from the Raspberry Pi to the robot through a server. Based on the received data, the robot will pick each object and place it in its corresponding, **designated location**, figure 3.

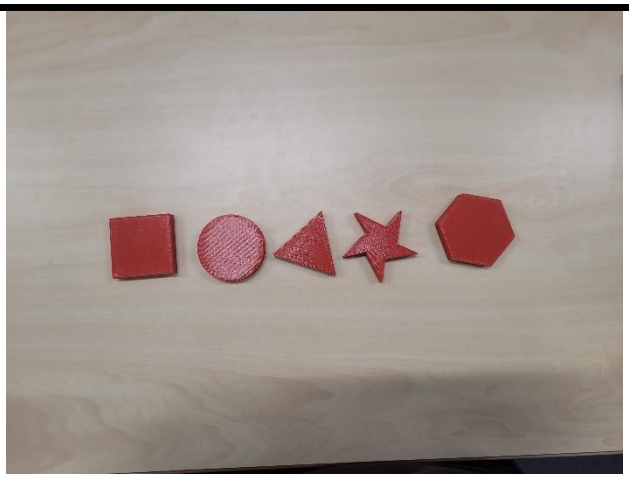


Figure 2 Object shapes for this project



Figure 3 Designated position for objects

Note:

- **Each student gets a circle, polygon or square, star or triangle.**
- **Students are required to train their own AI model to recognize three specific objects assigned by the instructor. The model must not detect or respond to the remaining two objects.** For example, student A shall detect Circle, Square and Star. The AI model should not detect Polygon and Triangle.
- **Example of a coding design principal is SOLID**

The project programs must include:

Raspberry Pi program Requirements:

Students are required to implement multithreading using, where each thread is responsible for handling at least one distinct functionality.

- **Basic AI model inference** – Running the trained AI model to detect object shape, position, and orientation.
- **Coordinate Translation** – Converting the object's pixel coordinates from the camera image into the robot's work object (world) coordinates.
- **Server communication** – Sending the processed data from the Raspberry Pi to the robot.
- **A graphical user interface (GUI)** – Displaying the detected object's position and orientation.

Robot Program Requirements:

- **Client Communication** – The robot must be able to communicate with the server to receive picking data.
- **Dynamic Picking** – The robot should dynamically pick objects based on detected positions provided by the AI model.
- **Fixed-Position Placing** – Each picked objects must be placed at a predefined, fixed location.

Prerequisite

The project is connected to previously solved instructions. If you have not done these projects, you will have a hard time.

Hand in and Examination

Examination will be done at the automationlab where you will have 3 hours to set up, start and check your programs, calibrate tools, make adjustments etc, and show the solution and present according to what grade you aim for.

Afterwards, students must submit a single ZIP file named FinalProject_StudentName.zip, which should include the complete program files, the trained AI model, a Pack-and-Go file of the robot station from the examination.

Note: Submissions that fail to follow the above-mentioned pattern will not be assessed and will be reported as failed. No resubmission is allowed in this case.

Grading

Students need to solve the project with grade level I to pass the project by grade E. To receive grade C, students need to solve the project with grade level III. To receive a grade of A, the Raspberry Pi program must demonstrate a well-designed architecture and follow object-oriented programming (OOP) principles.

Solved Exercises	Grade
Grade level I	E
Grade level II	D
Grade level III	C
Grade level III Object oriented coding	B-A

Grade Level I:

The submitted program must meet the following requirements:

- **Correctly Identify All Three Objects**
  The system must successfully detect and classify **all three** target objects.
- **Object Picking**
  All three objects must be picked up successfully. There is **no specific order** required for picking the objects.
- **Object Orientation and Placement**
  At least **two out of three** the picked objects must be placed with the correct orientation.
- **All code submitted by the student must also be commented and properly named.**

Grade Level II:

The submitted program must meet the following requirements:

- **Correctly Identify All Three Objects**
  The system must successfully detect and classify **all three** target objects.
- **Object Picking**
  All three objects must be picked successfully in each cycle. There is **no specific order** required for picking the objects.
- **Object Orientation and Placement**
  At least **four out of the six** objects must be placed with the correct orientation.
- **Color Detection**
  The system should be capable of **picking only the objects of a specified color**, based on the task or user input.

- **Continuous Execution**
  The new picking cycles should be executed without restarting the program.
    - **The program should be designed to support an infinite number of cycles;** two cycles will be tested.
    - **User input (e.g., button press or command) is required to trigger the next cycle.**
- **The AI inference time must not exceed 3 seconds, meaning the model should return detection values and results within this time.** The program must print the inference time.
- **All code submitted by the student must also be organized structure and properly named and commented.**

Grade Level III:

- **Object Picking**
  All three objects must be picked successfully in each cycle.
- **Object Orientation and Placement**
  At least 5 out of the six objects (at least one of each shape) must be placed with the correct orientation.
- **Color Detection**
  The system should be capable of **picking only the objects of a specified color**, based on the task or user input.
- **Continuous Execution**
  The new picking cycles should be executed without restarting the program.
    - The system should be designed to handle an **infinite number of picking cycles.**
    - **User input** must be used to **select which object** to pick in each cycle**.**
- Students are allowed to **retry picking each shape up to two times** if the initial attempt fails.
- **The AI inference time must not exceed 3 seconds, meaning the model should return detection values and results within this time.** The program must print the inference time.
- **All code submitted by the student must also be well-organized structure, properly named and commented.**